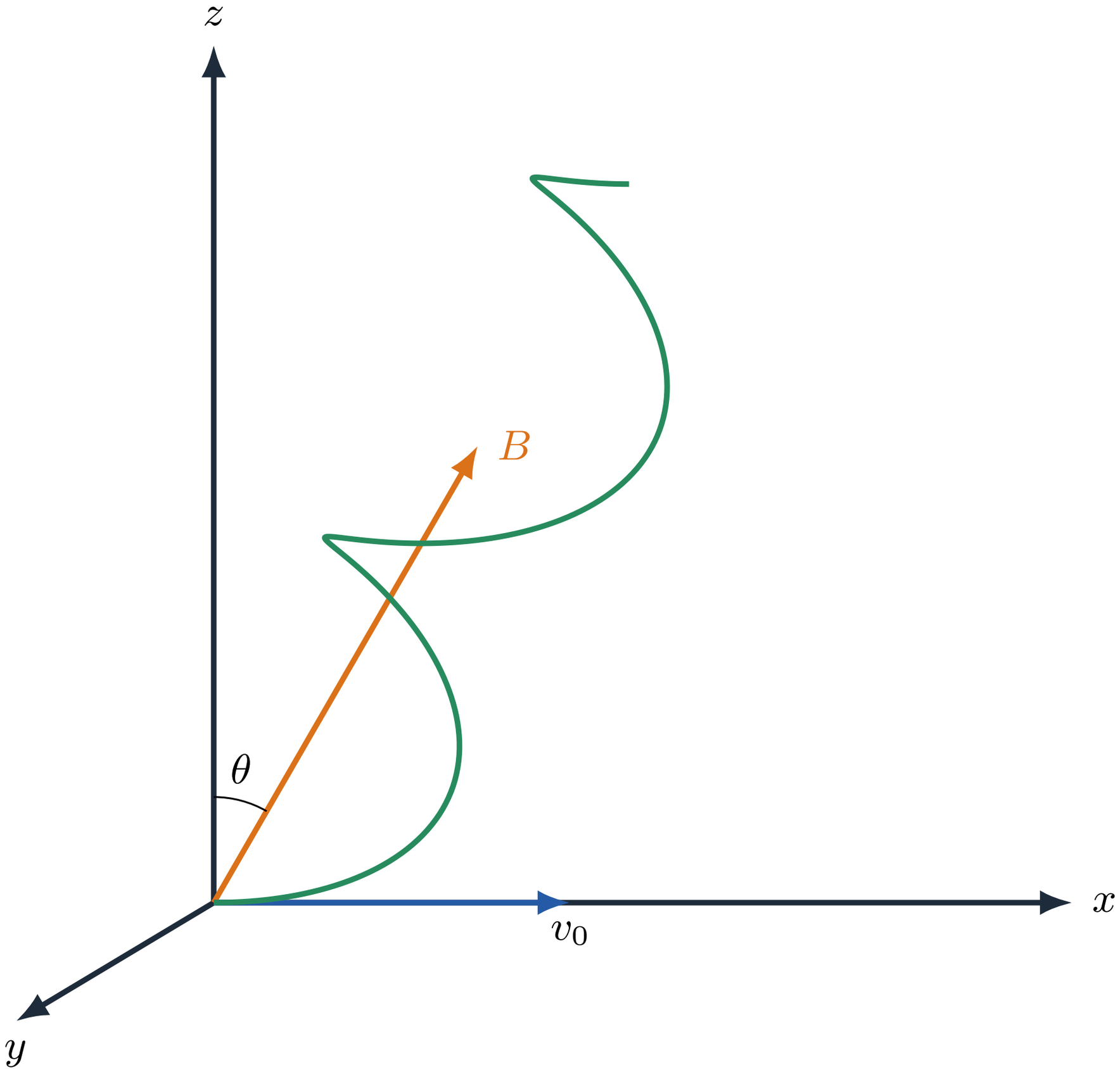
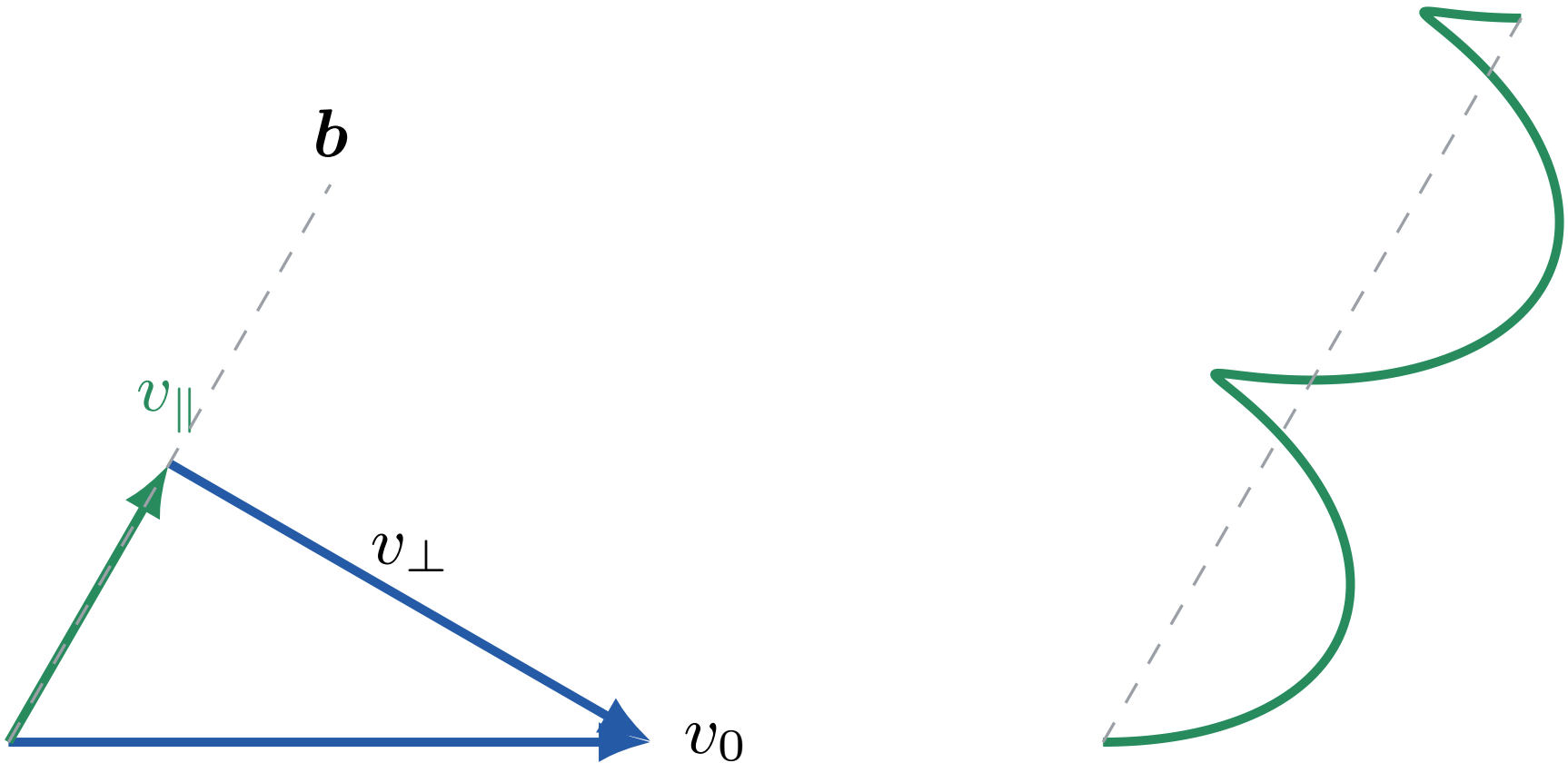
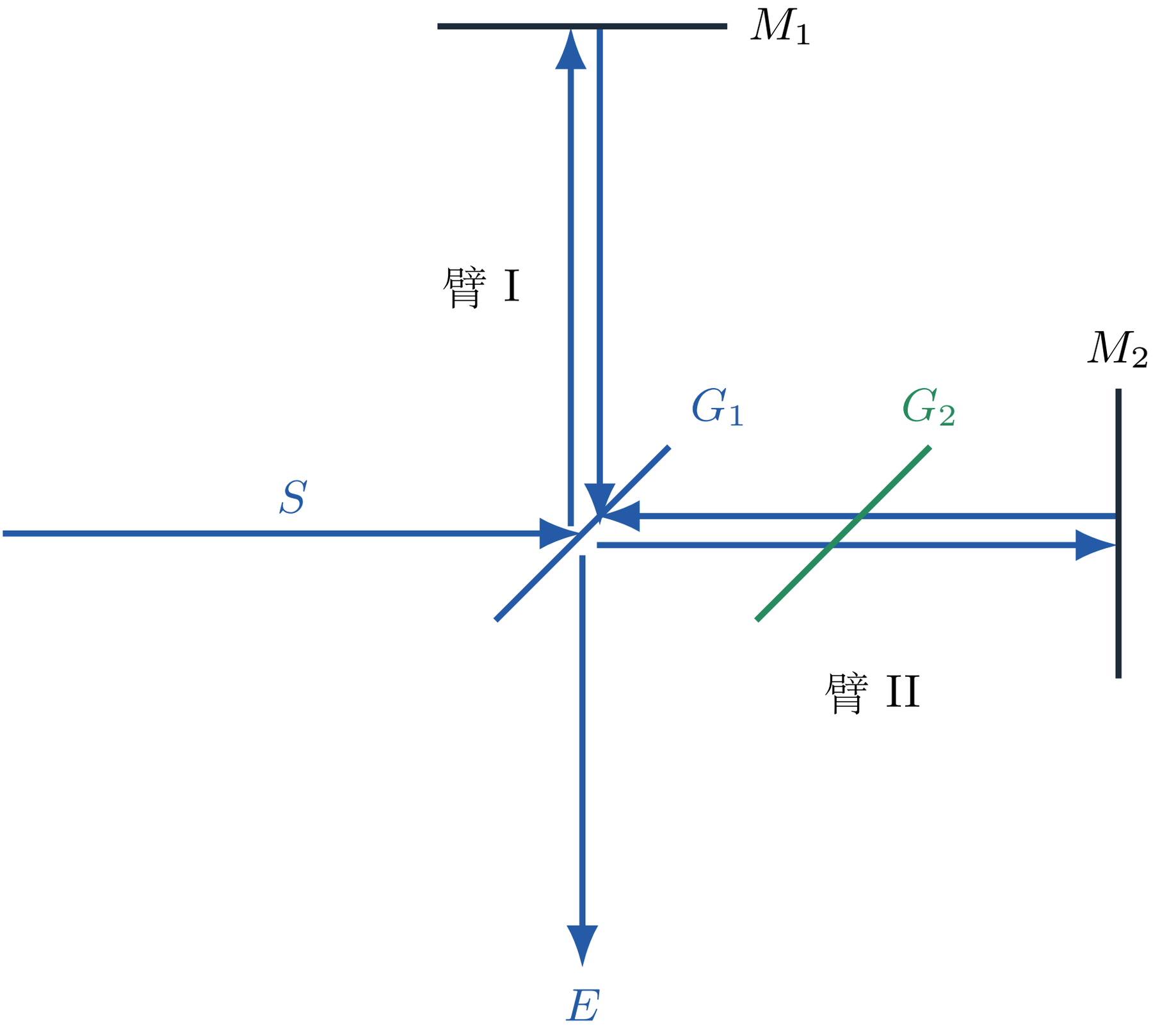


解析参数轨迹的斜投影：$\theta = 30°, q > 0$



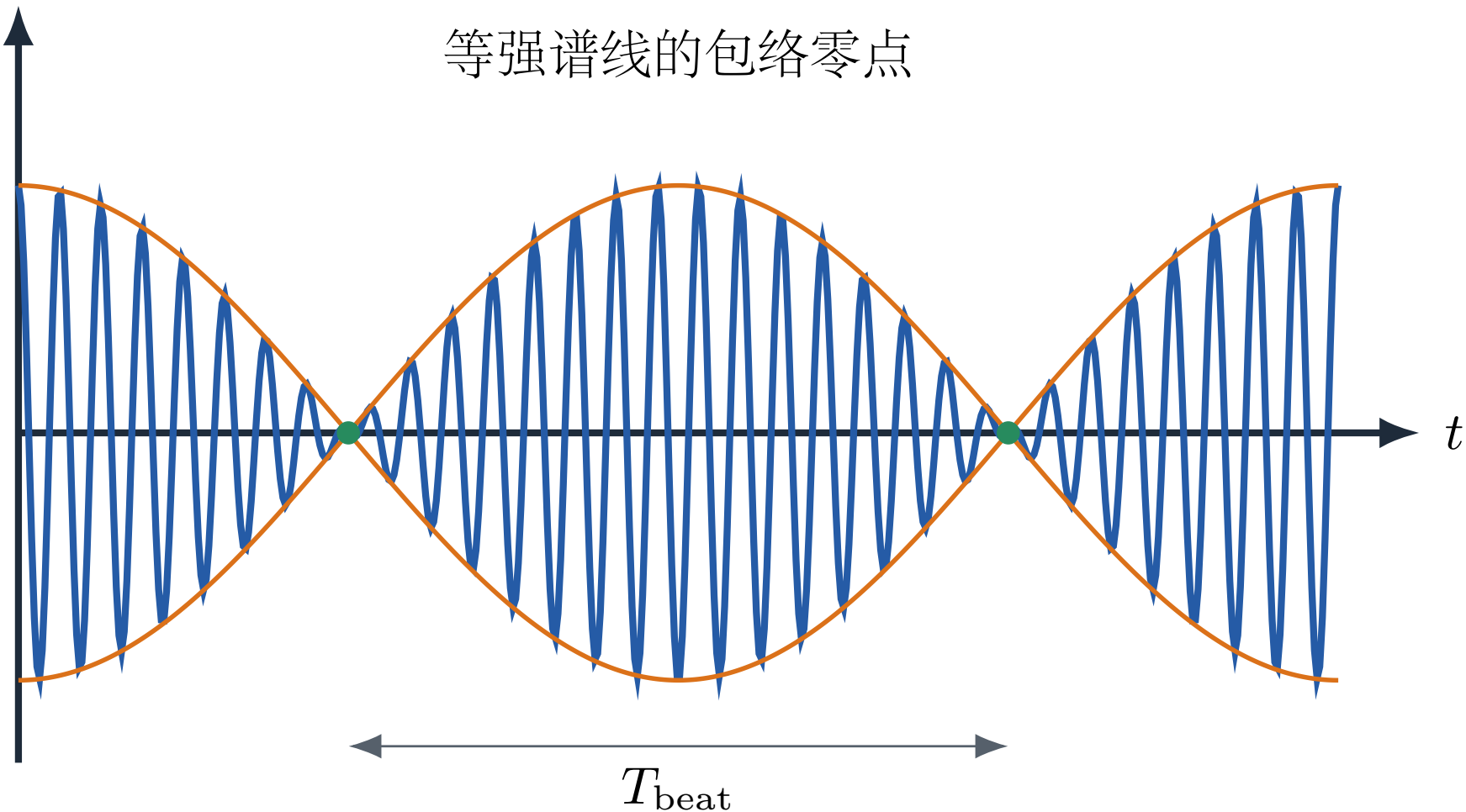
$$v_\parallel = v_0 \sin\theta, \quad v_\perp = v_0 \cos\theta$$

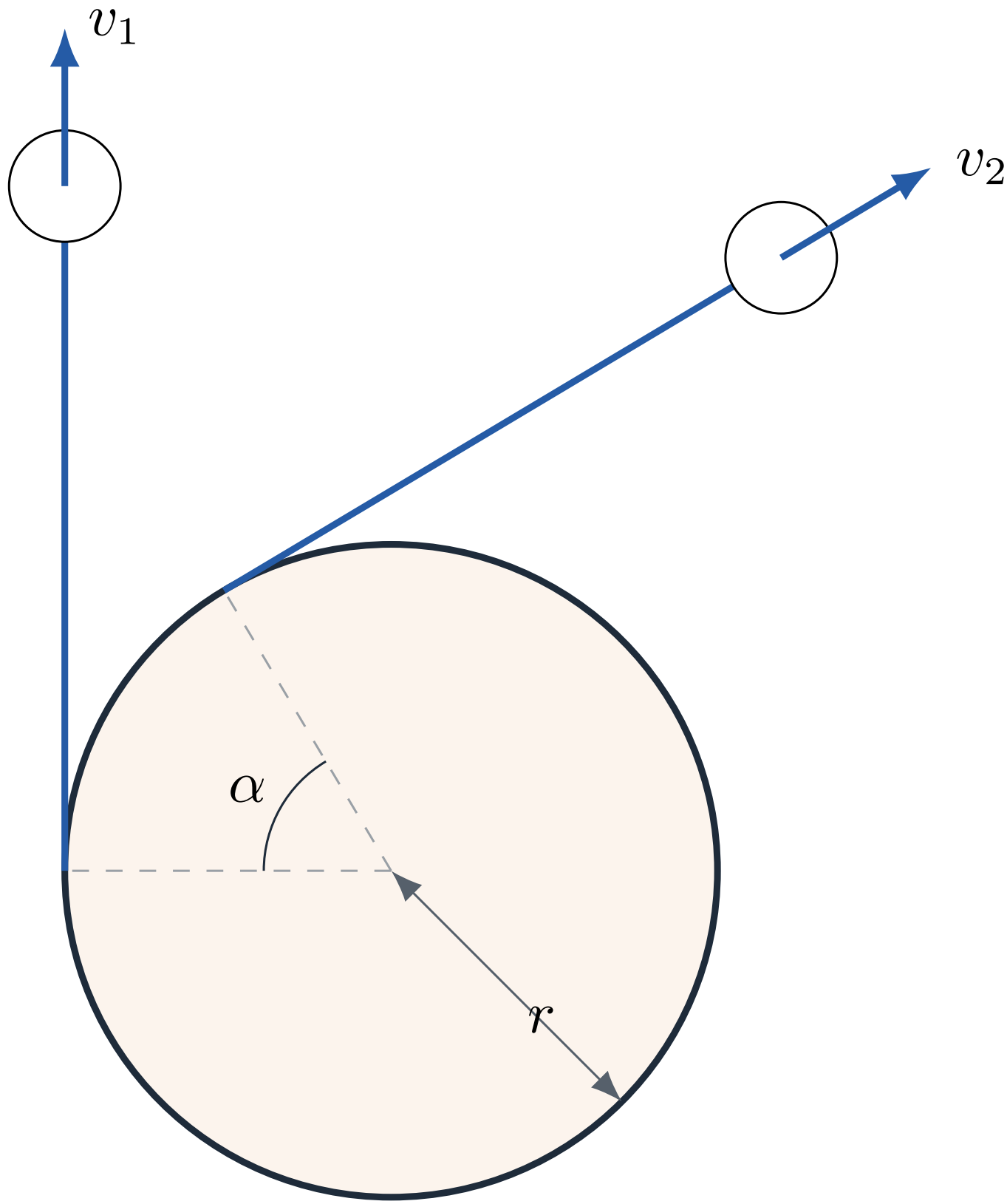
$$y = (v_0 \cos\theta/\Omega)(\cos\Omega t - 1)$$



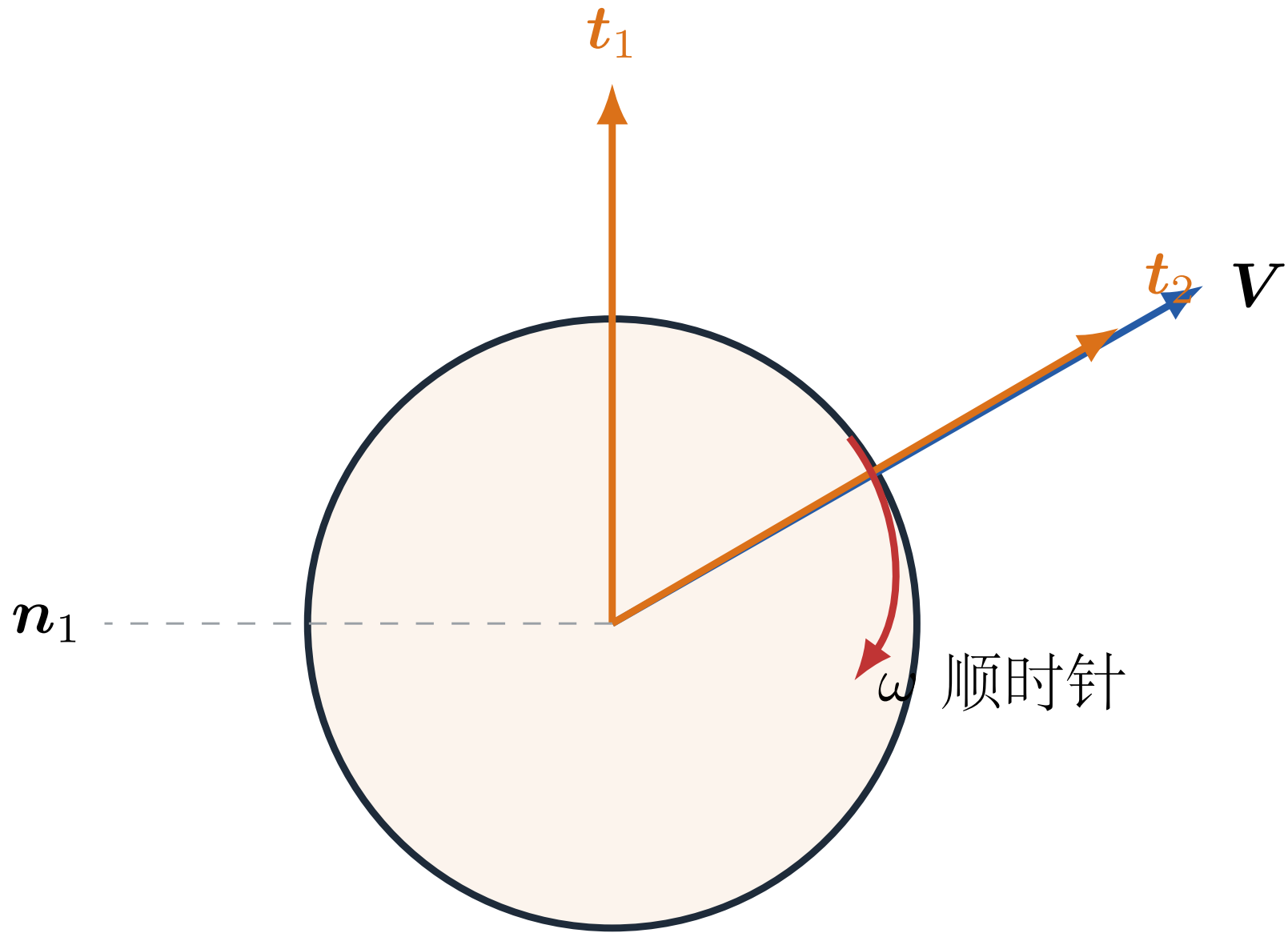
补偿板在右镜臂；两臂均标出往返

干涉调制





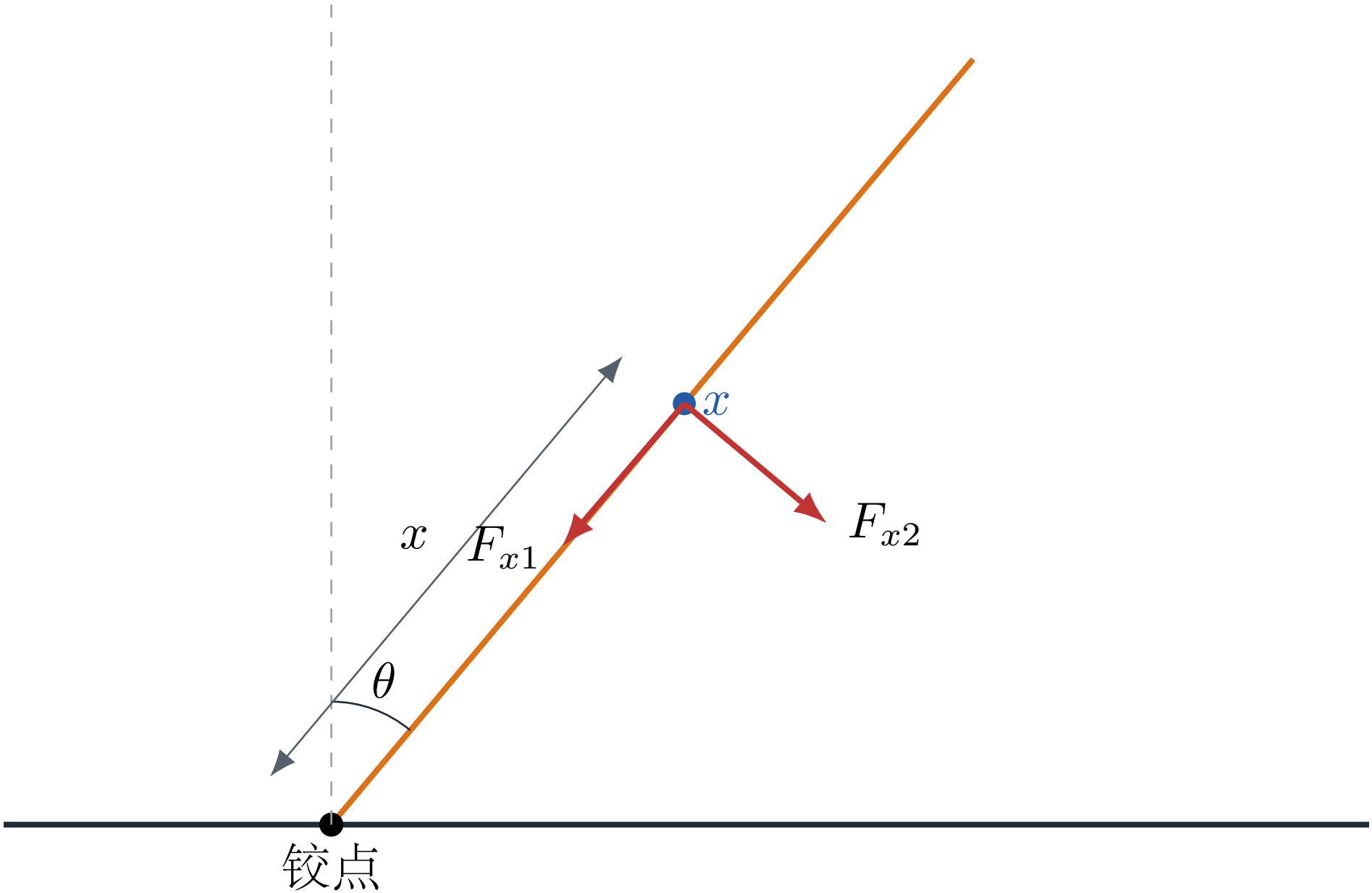
两条不可伸长绳分别沿圆柱的两条切线离开

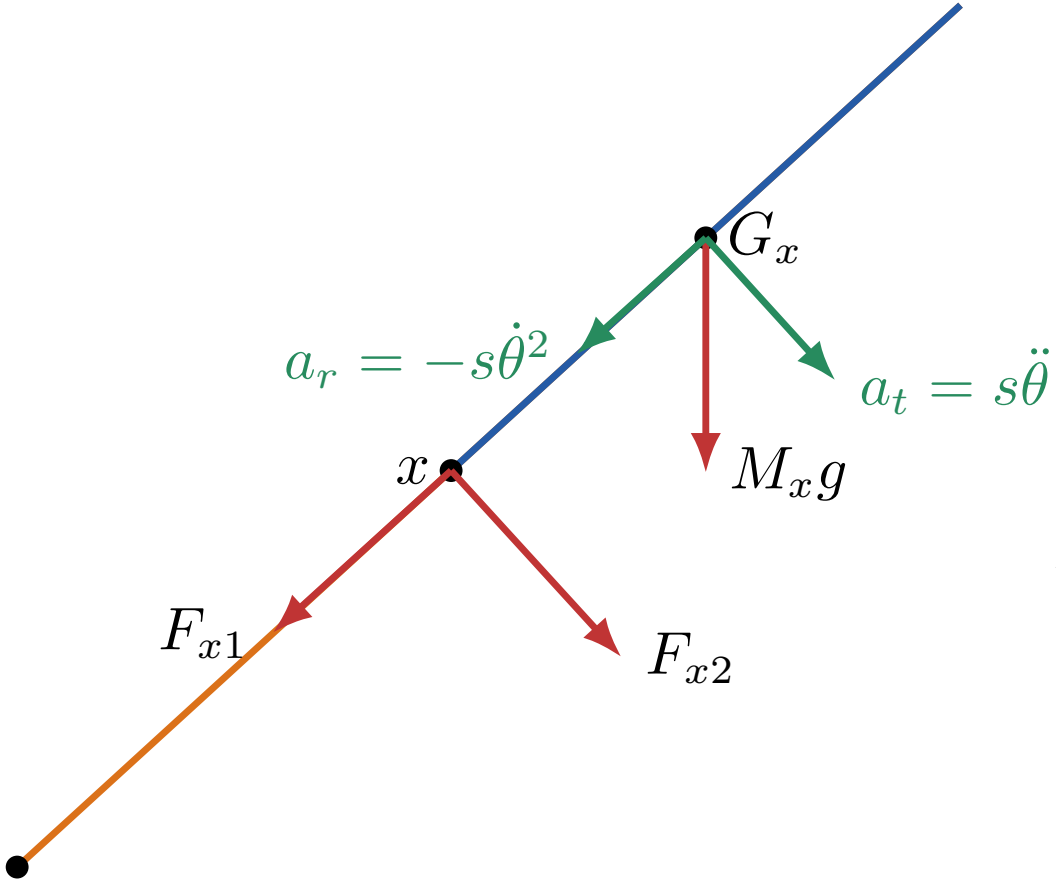


$$\boldsymbol{V} \cdot \boldsymbol{t}_i = v_i - r\omega \equiv u_i$$

$$V = \frac{\sqrt{u_1^2 + u_2^2 - 2u_1 u_2 \cos\alpha}}{\sin\alpha}$$

$$\omega_1 = -\frac{\boldsymbol{V} \cdot \boldsymbol{n}_1}{l_1}$$

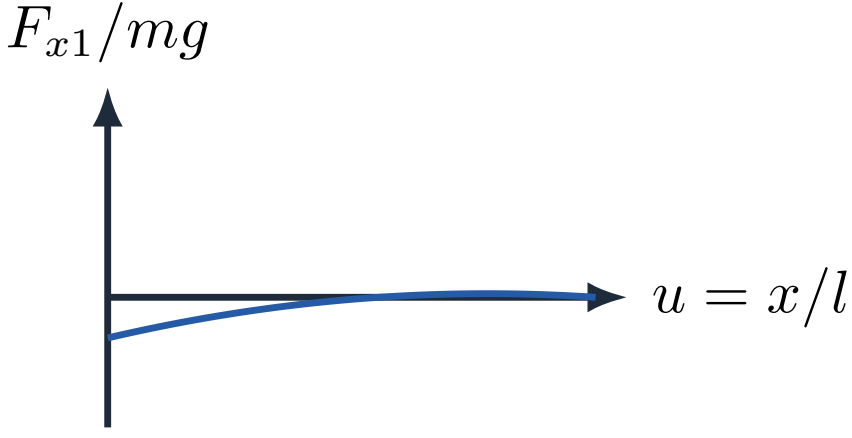

x
x
F_{x1}
F_{x2}
θ
铰点



$$M_x = m(l-x)/l$$

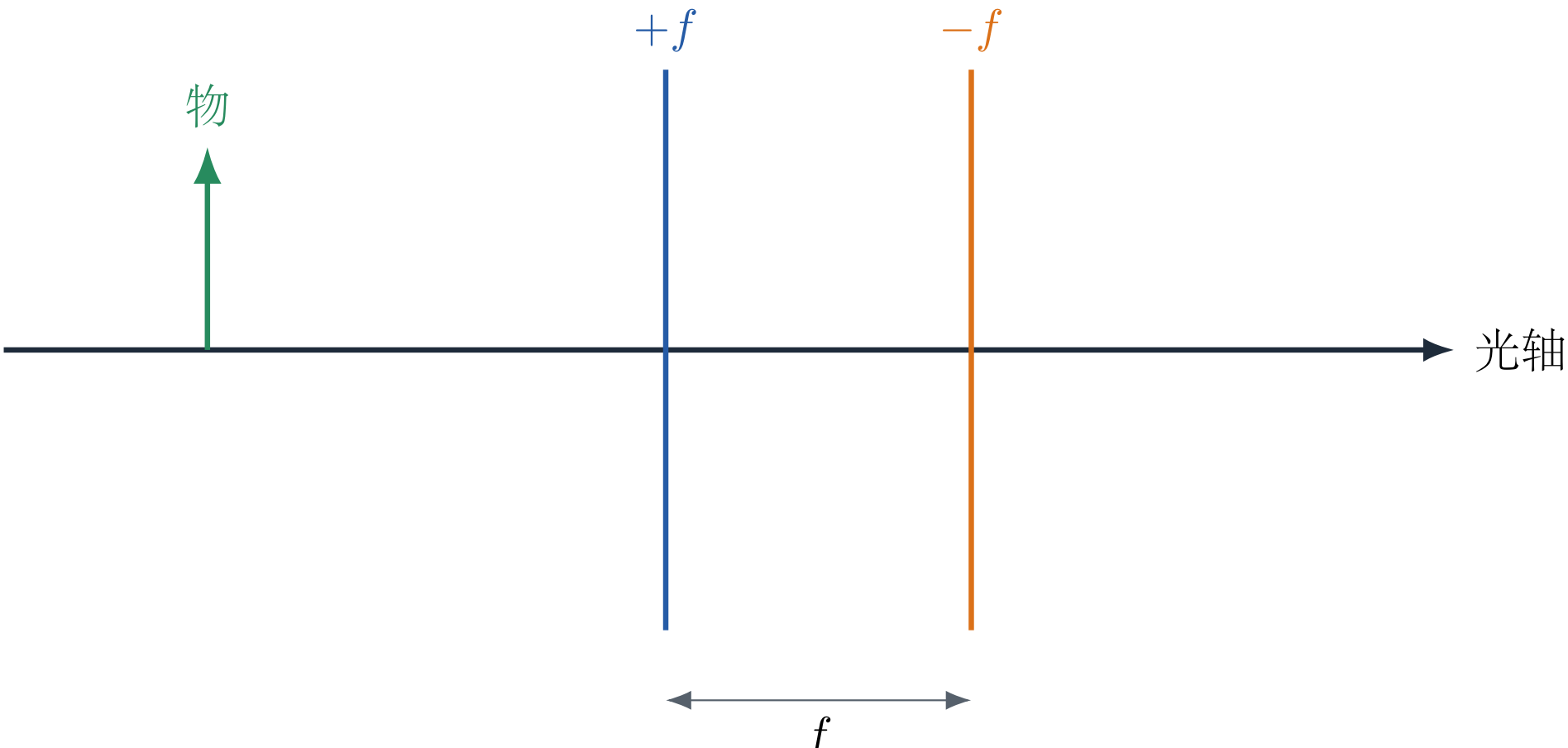
$$s = (l+x)/2$$

$$\dot{\theta}^2 = 3g(1-\cos\theta)/l$$

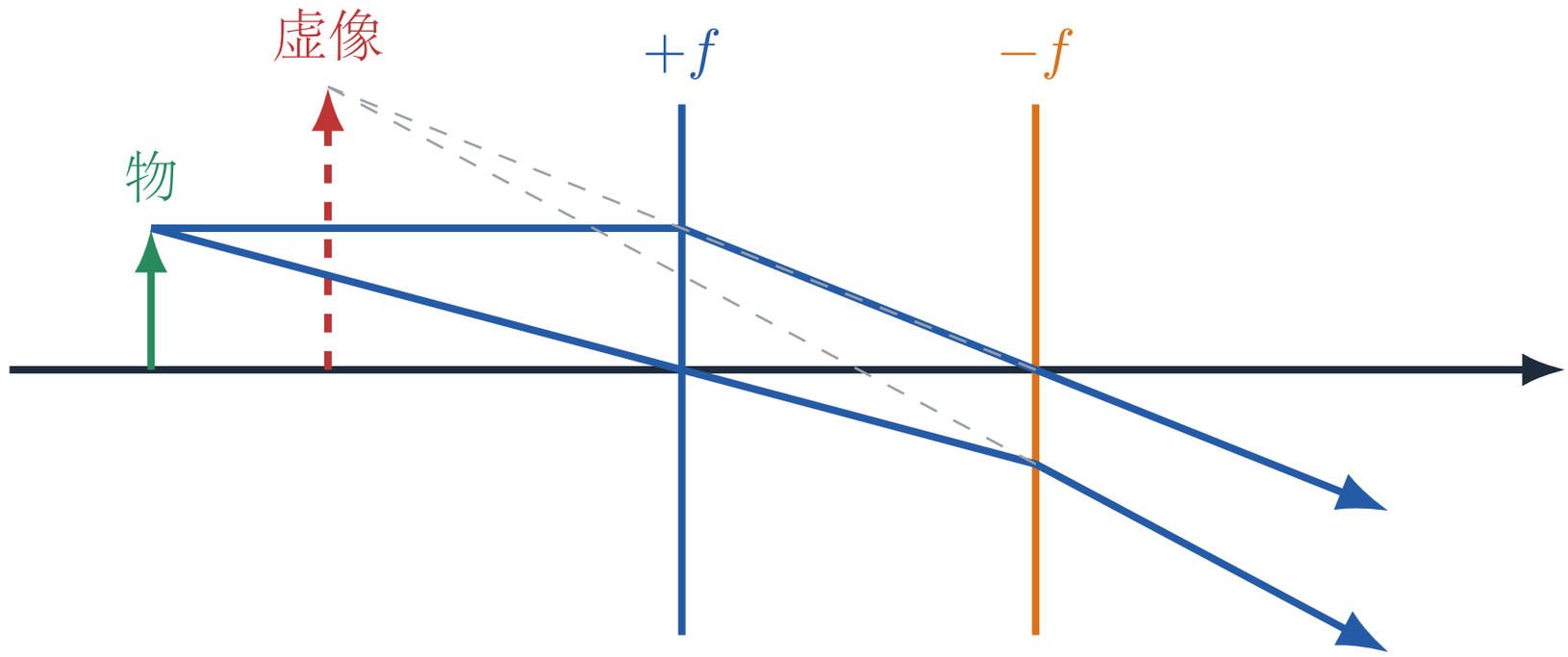


$c = 0.7$：根部受压

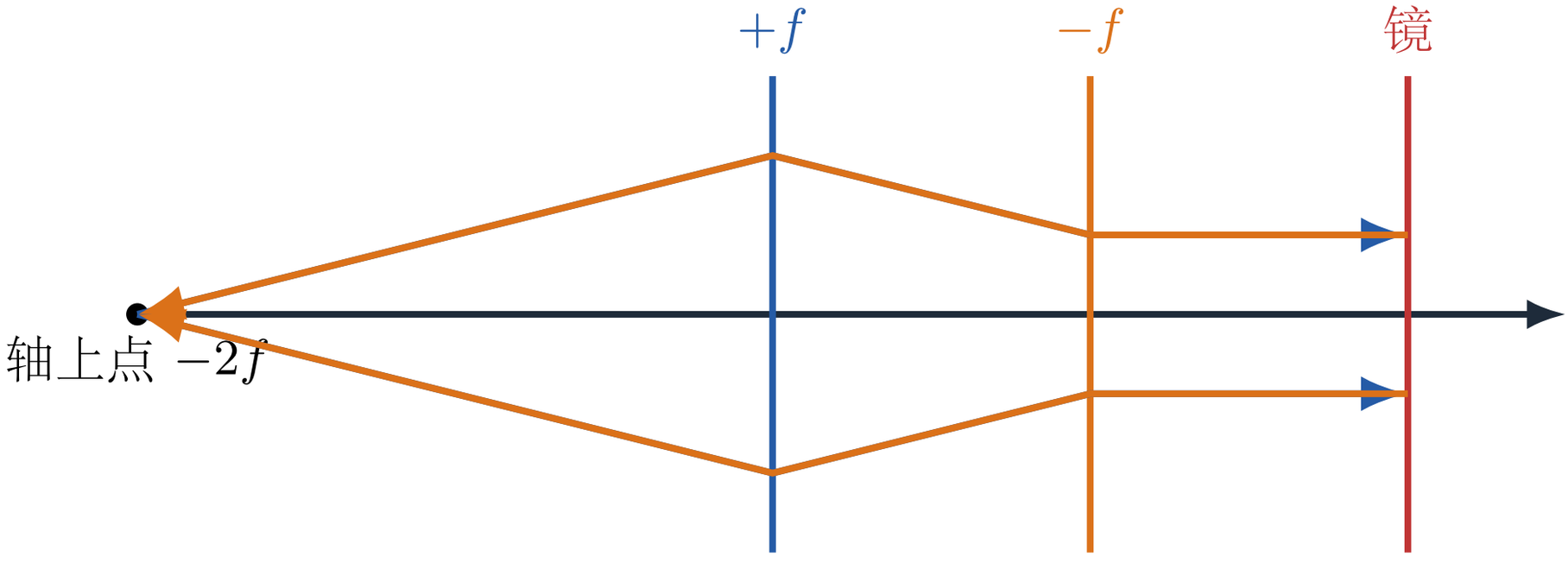
自由端附近先受拉



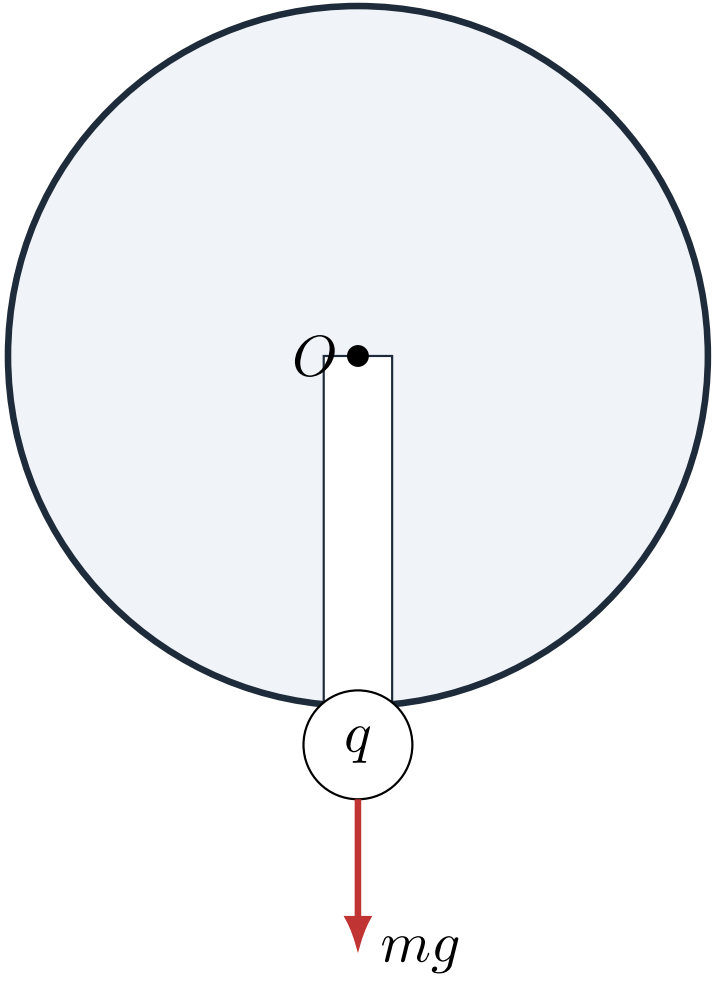
焦距分别为 f、$-f$ 的薄透镜相距 f



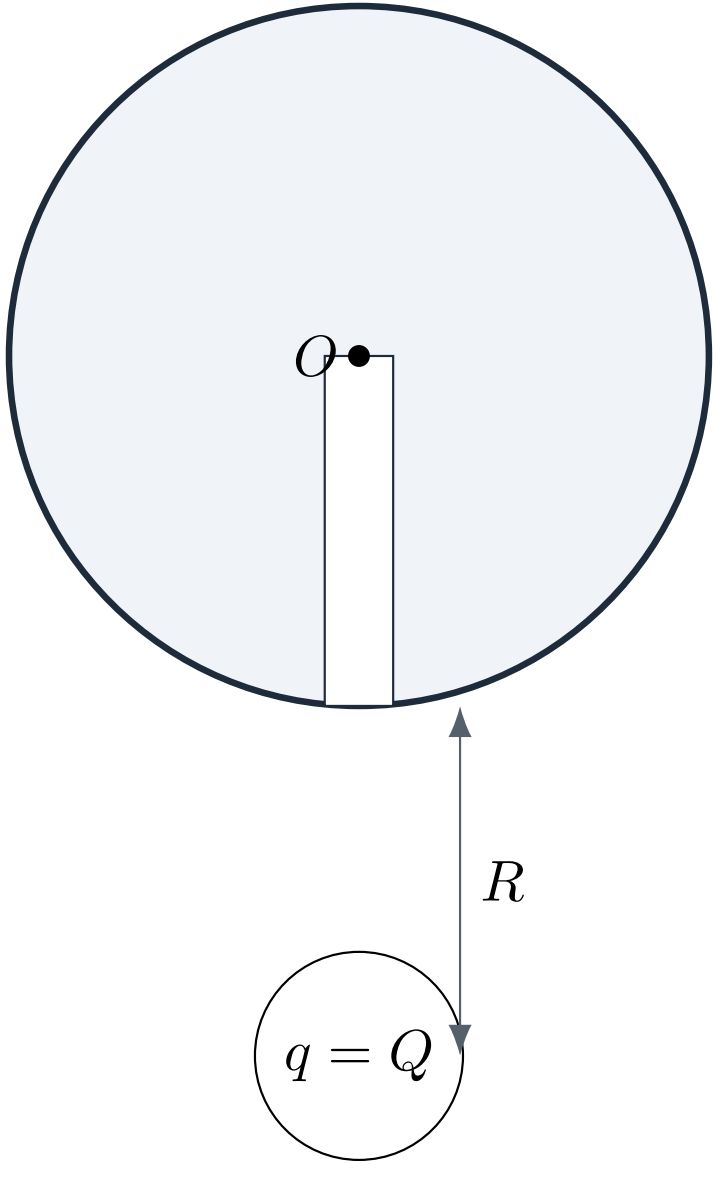
$a = 1.5$：$s_2' = -2f$，全局像坐标 $-f$



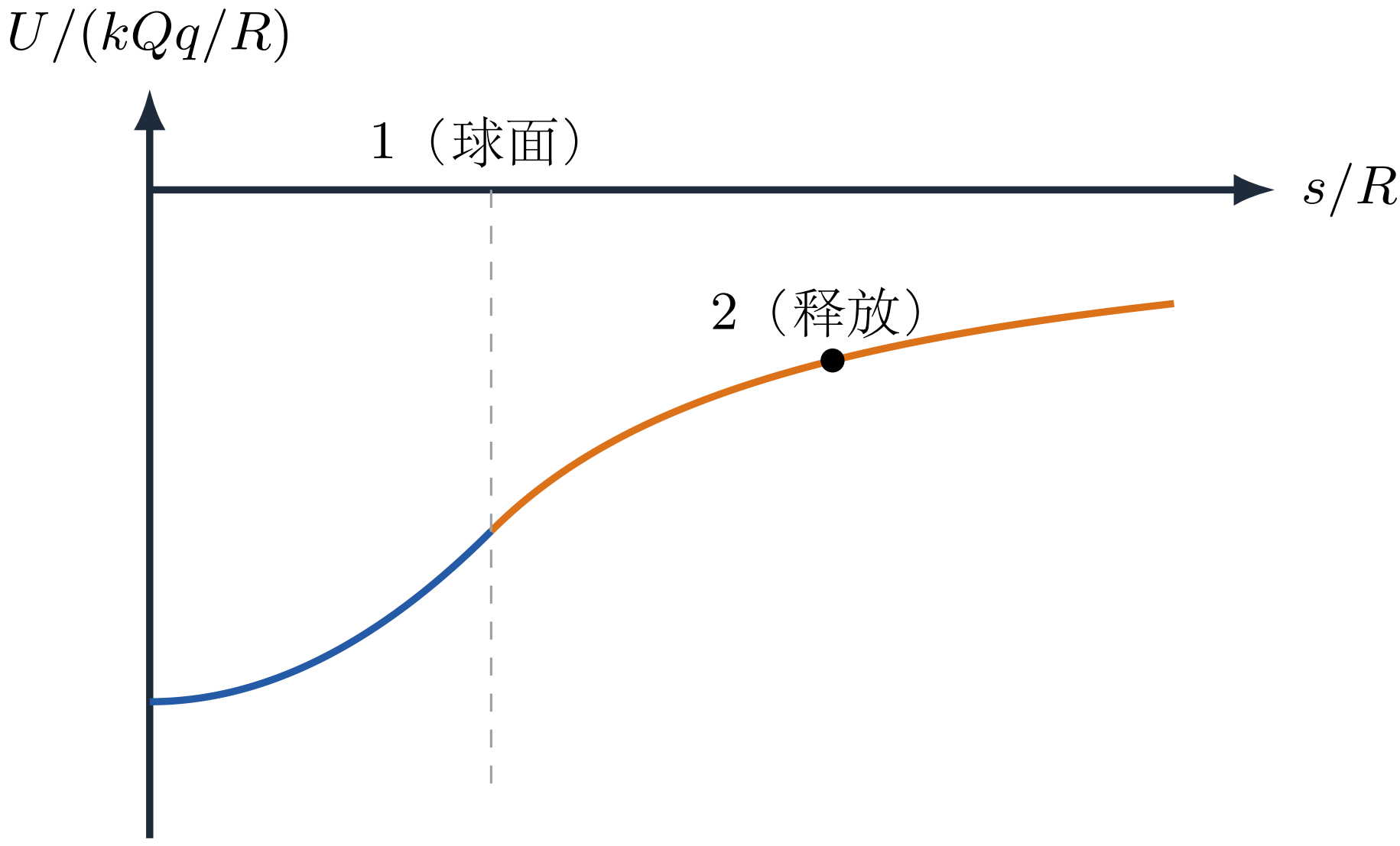
轴上点的平行光往返，镜距任意



球面释放，考虑重力



距球面 R 处释放，忽略重力



球内二次势与球外 $-1/s$ 势连续相接

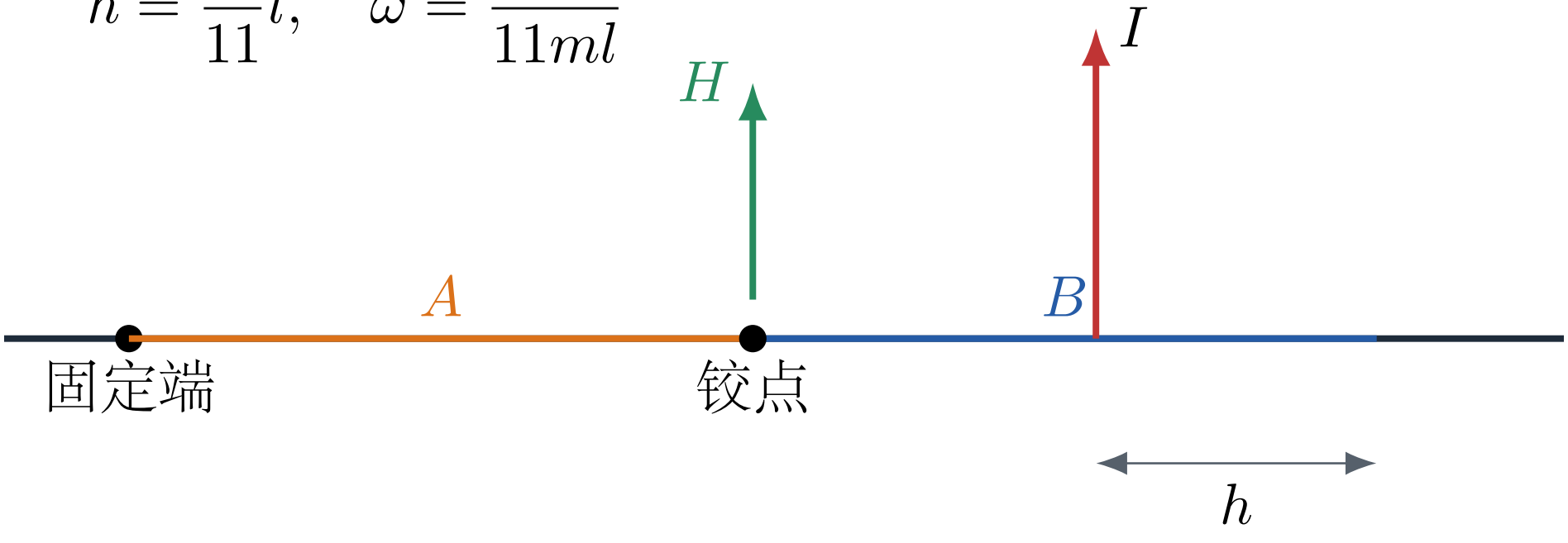


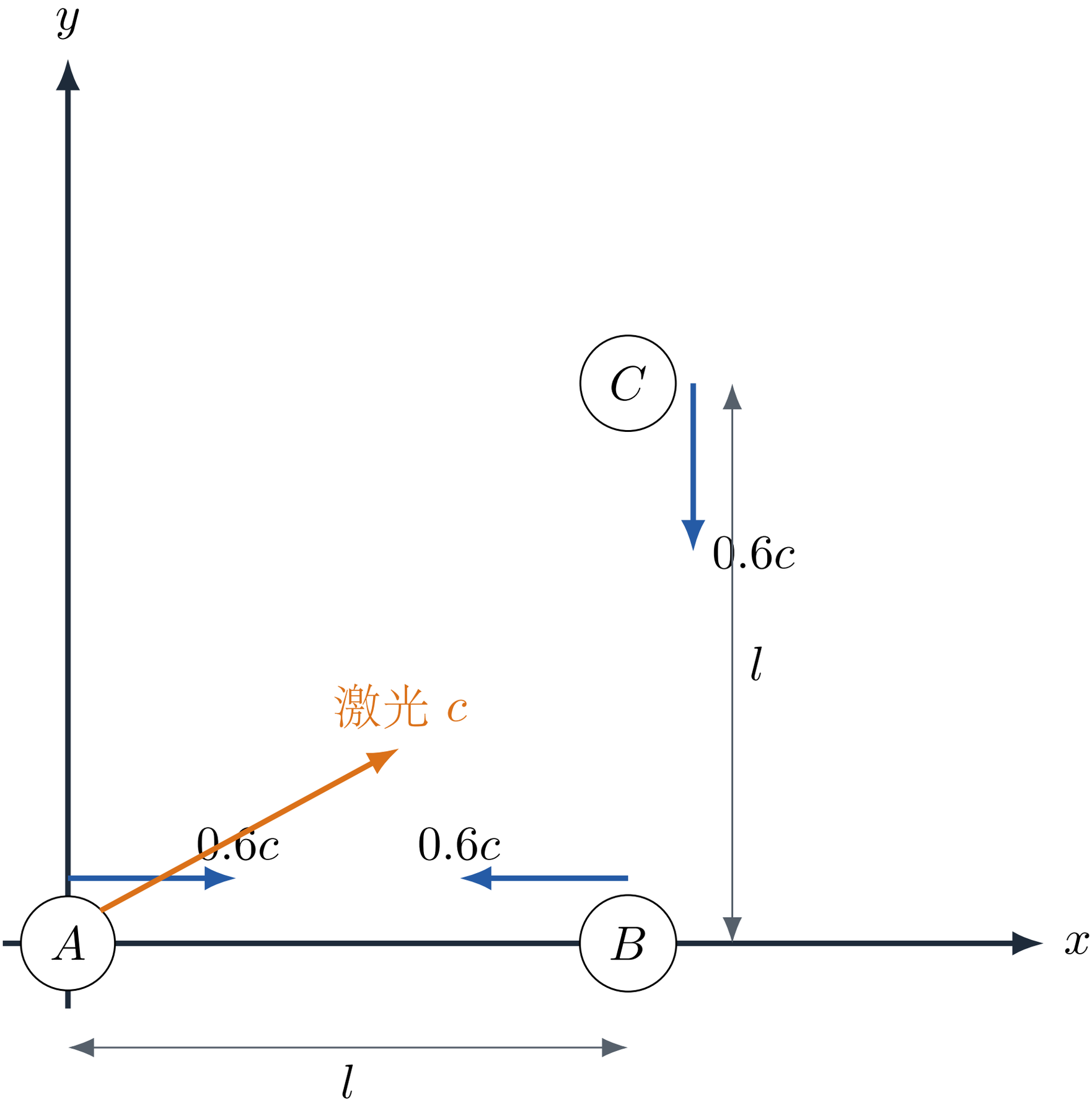
水平虚线为释放姿态；实线为下落后的姿态

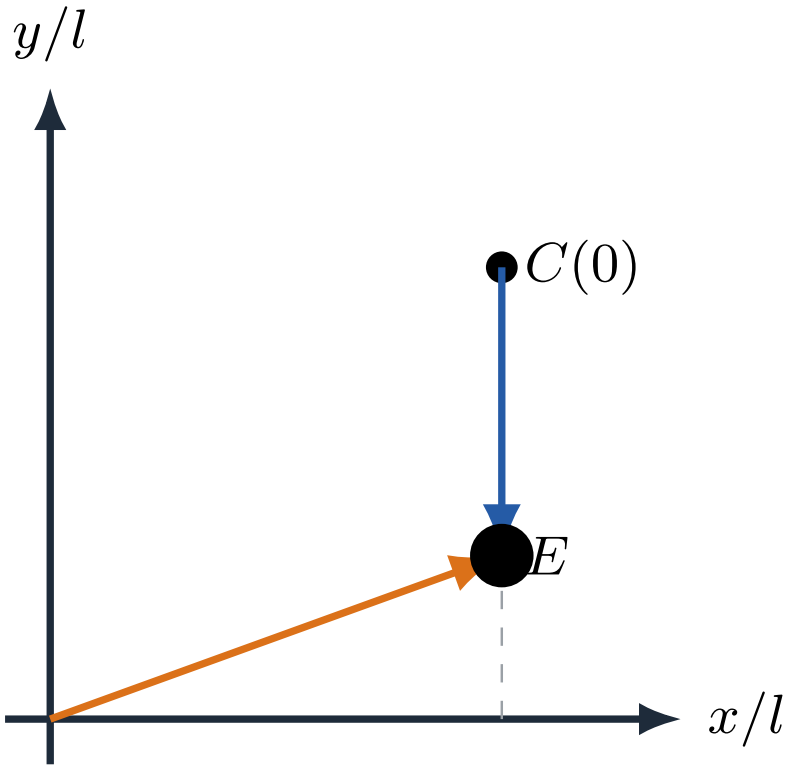
$$I + H = \frac{3}{2} ml\omega$$

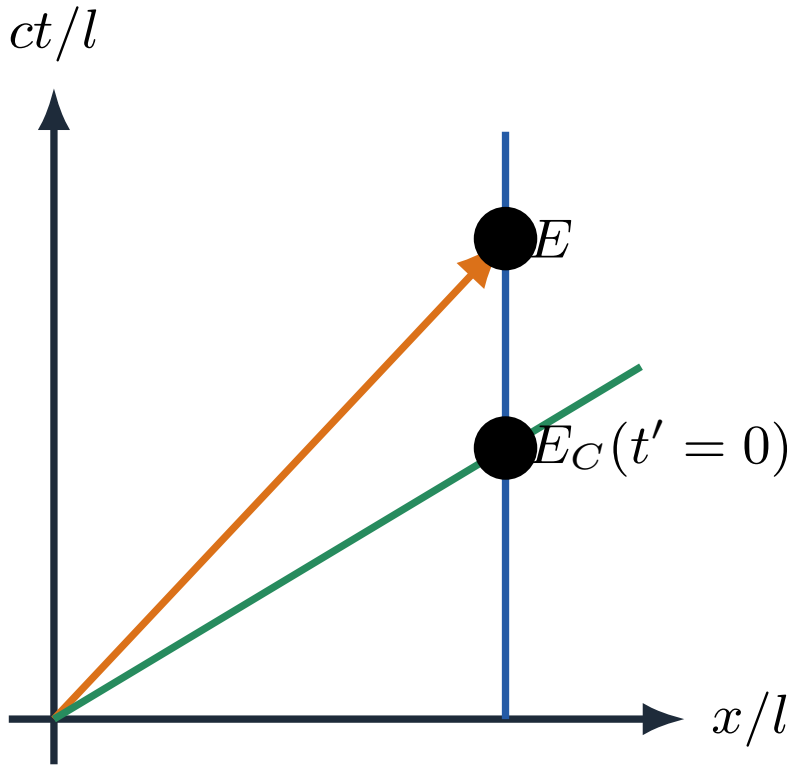
$$-Hl = \frac{1}{3} ml^2\omega$$

$$h = \frac{6}{11} l, \quad \omega = \frac{6I}{11ml}$$




y
C
0.6c
l
激光 c
0.6c
0.6c
A
B
x
l





$cT/l = \sqrt{1 + (1 - 0.6cT/l)^2}$

绿色：$ct = 0.6x$ 同时截面